TECHNISCHE HOCHSCHULE DEGGENDORF
Technologie Campus Cham

Predictive Maintenance Analysis

Presentation Agenda

01

Introduction & Dataset

Project goals, predictive maintenance basics, and raw data overview.

02

Exploratory Data Analysis

Visualizing correlations, sensor distributions, and failure modes.

03

Data Preparation & Modeling

Feature engineering, SMOTE balancing, and model selection.

04

Results & Business Impact

Performance metrics, comparison, and practical applications.

Project Team

❖ Group (3)

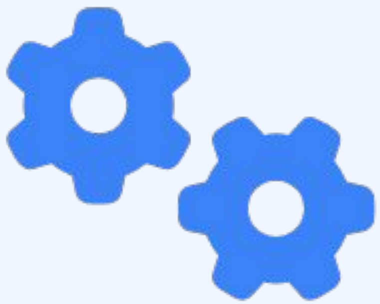
Mohamed Elmekawy	Introduction & Dataset
Mahmoud Elkfafy	Exploratory Data Analysis
Talaat Sallam	Data Preparation & Modeling
Eman Gabr	Results & Business Impact

Section 1: Introduction & Dataset

Predictive Maintenance (PdM)

Predictive maintenance uses data-driven, proactive methods to evaluate the condition of equipment and predict when maintenance should be performed.

- Reduces unplanned downtime
- Optimizes maintenance schedules
- Maximizes tool life and resource usage
- Improves overall operational safety



Project Problem Statement

Industrial machines often fail unexpectedly, leading to significant economic losses. The challenge is to distinguish between normal operation and impending failure using real-time sensor data.

Objectives:

- Analyze the AI4I 2020 synthetic dataset for predictive patterns.
- Address the severe class imbalance (only 3.39% failure rate).
- Develop an ML model to predict failure with high recall to avoid missing critical issues.

Dataset Overview

The AI4I 2020 Predictive Maintenance Dataset contains 10,000 observations representing industrial processes.

Attribute	Value
Total Samples	10,000
Total Features	14 Columns
Failure Samples	339 (3.39%)
Non-Failure Samples	9,661 (96.61%)
Data Integrity	0 Missing Values

Predictive Maintenance Parameters

Input Sensor Features

Thermal

- Air Temperature [K]
  - Process Temperature [K]

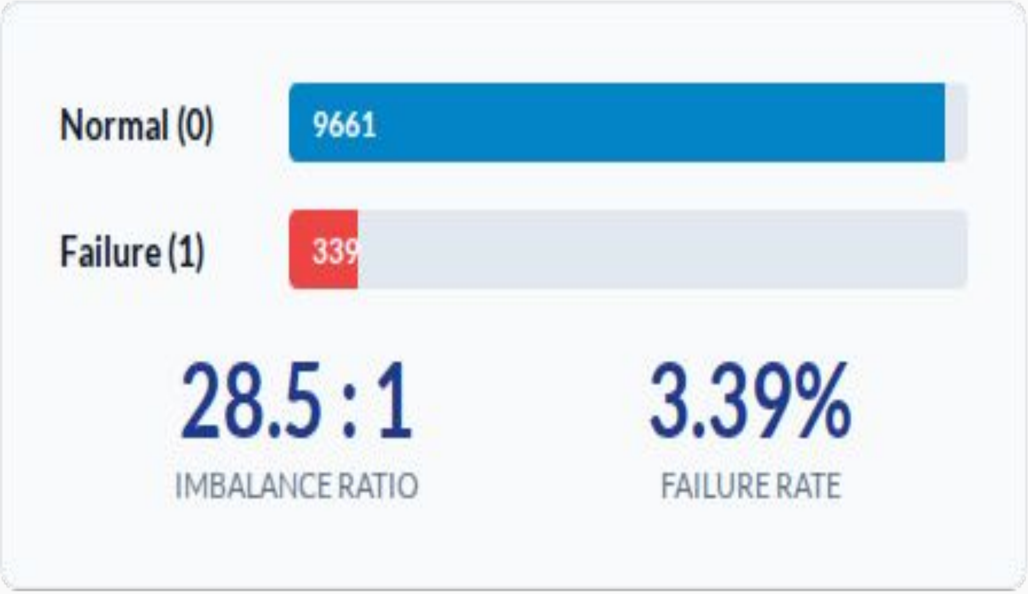
Mechanical

- Rotational Speed [rpm]
- Torque [Nm]

Structural & Categorical

- **Tool Wear:** Usage time [min]
- **Type:** Quality Level (L, M, H)

Target Variable: Failure



Section 2: Exploratory Data Analysis

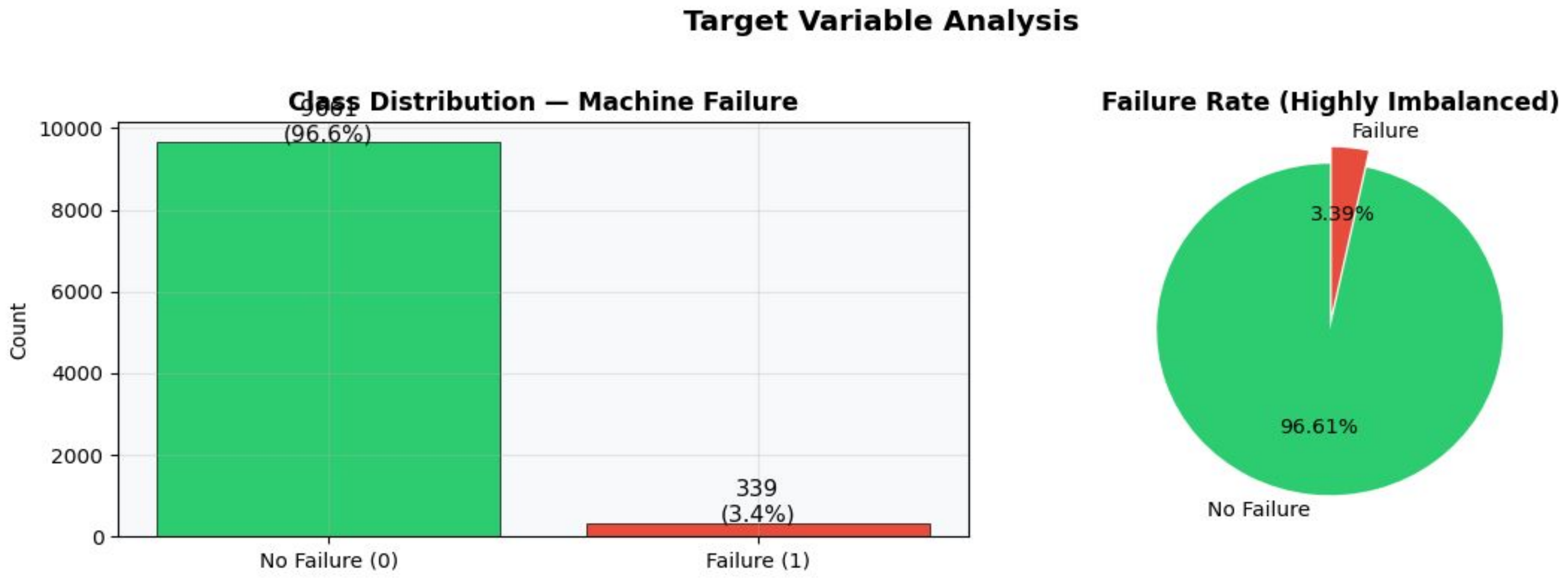
1- Dataset Overview

ML models cannot learn properly if:

- The dataset is highly imbalanced (**9661** normal vs **339** failures).
- Data is small

Observation: Class imbalance ratio is **28.5:1**.

So we use **SMOTE** stands for: **Synthetic Minority Over-sampling Technique**



2- Checking Missing Values and Data Types

ML models cannot learn properly if:

- many values are missing
- features are corrupted

```
=== Missing Values ===
UDI                        0
Product ID                 0
Type                       0
Air temperature [K]        0
Process temperature [K]    0
Rotational speed [rpm]     0
Torque [Nm]                0
Tool wear [min]            0
Machine failure            0
TWF                        0
HDF                        0
PWF                        0
OSF                        0
RNF                        0
dtype: int64
```

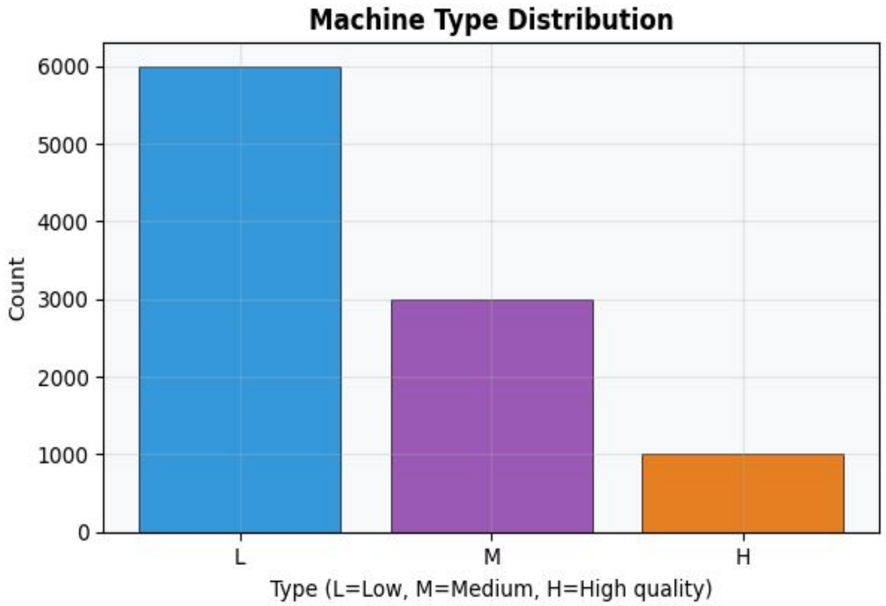
```
=== Statistical Summary ===
UDI                        | dtype: int64     | unique: 10000
Product ID                 | dtype: object    | unique: 10000
Type                       | dtype: object    | unique: 3
Air temperature [K]        | dtype: float64   | unique: 93
Process temperature [K]    | dtype: float64   | unique: 82
Rotational speed [rpm]     | dtype: int64     | unique: 941
Torque [Nm]                | dtype: float64   | unique: 577
Tool wear [min]            | dtype: int64     | unique: 246
Machine failure            | dtype: int64     | unique: 2
TWF                        | dtype: int64     | unique: 2
HDF                        | dtype: int64     | unique: 2
PWF                        | dtype: int64     | unique: 2
OSF                        | dtype: int64     | unique: 2
RNF                        | dtype: int64     | unique: 2
```

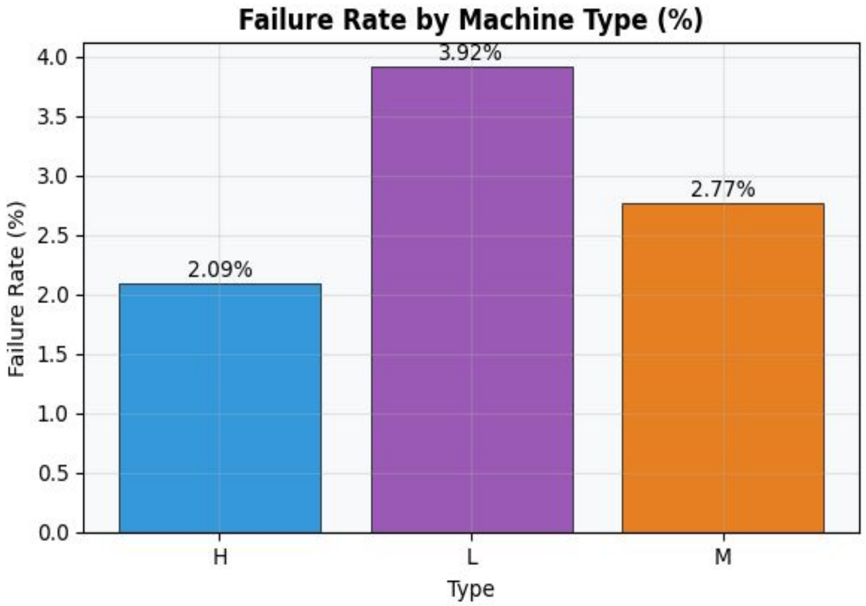
3- Machine Type Analysis

Distribution of Machine Types (L, M, H) and their associated failure rates:

Quality Type	Count	Failures	Failure Rate
Low (L)	6000	235	3.92%
Medium (M)	2997	83	2.77%
High (H)	1003	21	2.09%

Observation: Low-quality machine types (L) exhibit nearly double the failure rate of High-quality (H) machines, suggesting quality level is a significant predictor.

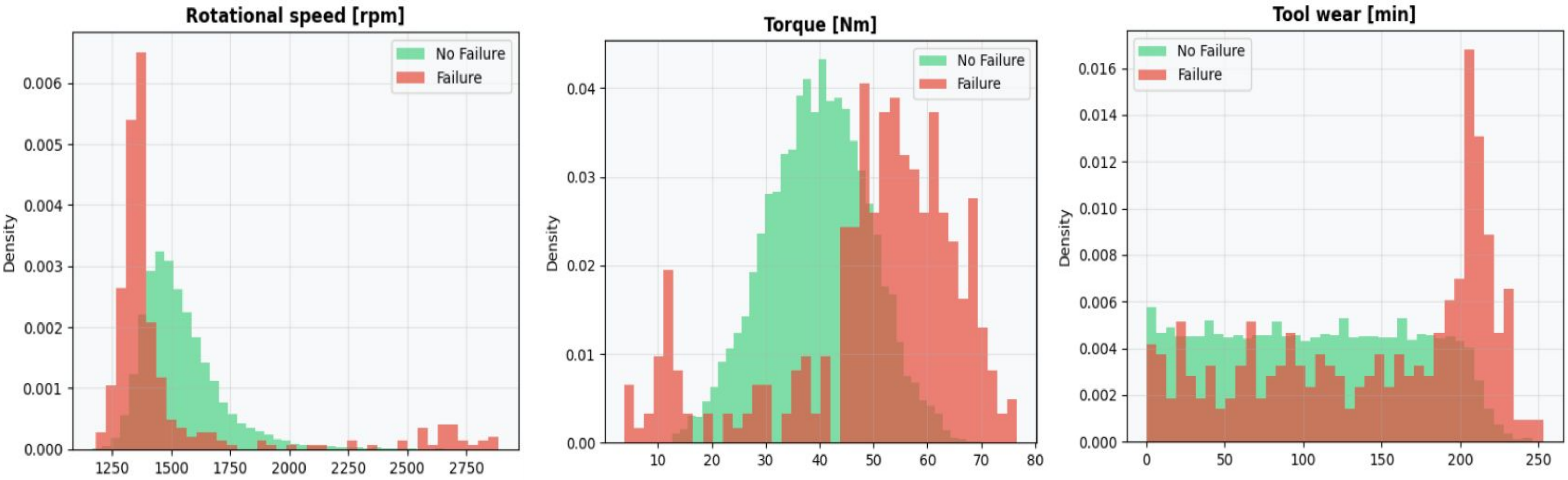




4- Sensor Reading Distributions

Analyzing how sensor values differ between Failure and No Failure states:

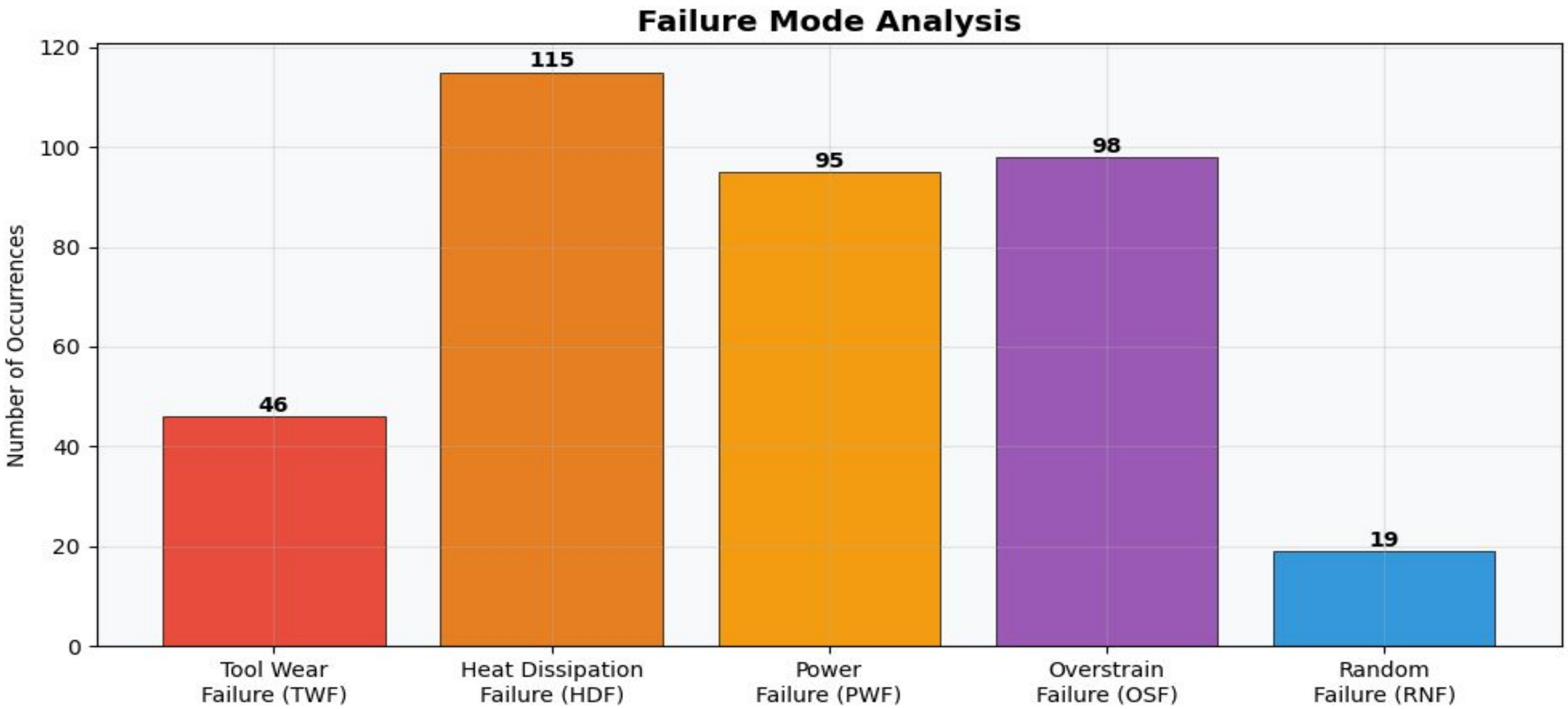
- **Tool Wear:** Failures are more common as tool wear approaches 200 minutes.
- **Torque:** Higher torque values (>60 Nm) correlate strongly with failures.
- **Rotational Speed:** Lower speeds generally associated with higher failure frequency.



5- Analysis of Failure Modes

Five specific failure types were recorded in the dataset:

Observation: Most failures arise from **thermal and mechanical issues** rather than random events, indicating that <u>**sensor measurements can effectively predict failures.**</u>

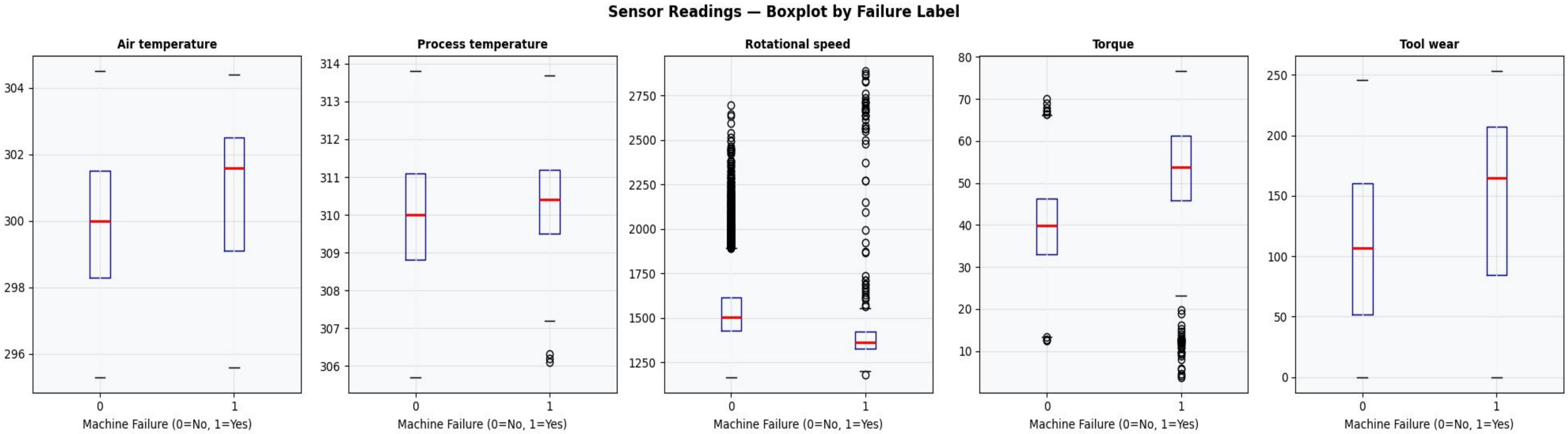


6- Outlier & Variance Analysis

Boxplots show that **failure machines** have **more variation and more unusual sensor values** than normal machines

Key Insight: In predictive maintenance, unusual sensor readings may indicate developing faults, making outliers valuable for failure prediction rather than noise to be removed.

- Failure machines tend to operate at lower rotational speeds and higher torque values.
- Tool wear is significantly higher in failure cases.
- Several outliers are observed, particularly in rotational speed and torque.
- These outliers were retained because they represent valid abnormal operating conditions rather than data errors.



Key EDA Insights

Quality Matters

Machine Type 'L' is more failure-prone than 'H'.

The Speed-Torque Tradeoff

High torque at low speed is a primary failure indicator.

Multi-Failure

Some failures involve multiple modes simultaneously.

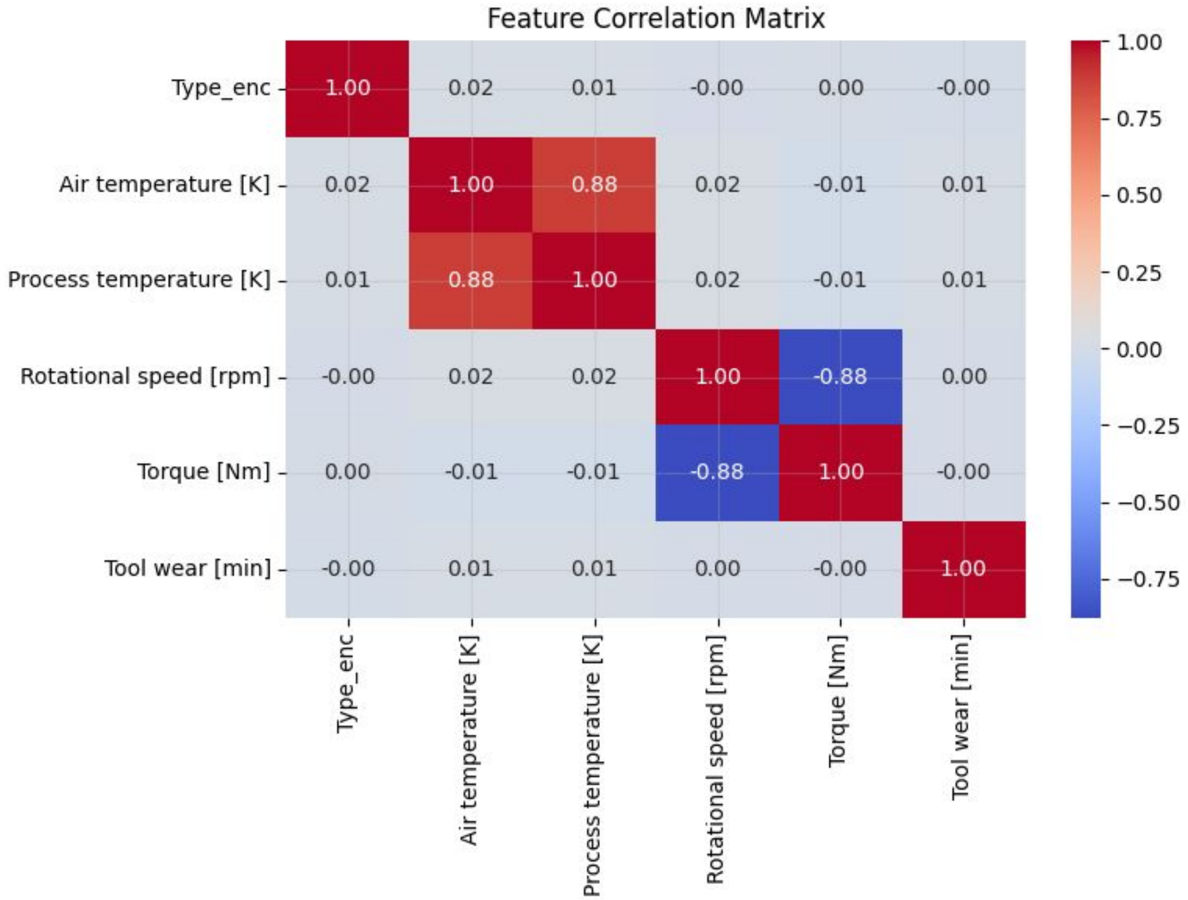
Section 3: Data Preparation & Modeling

Feature Engineering Strategy

Correlation Analysis

Observation

- Strong correlation between Air Temperature and Process Temperature.
- Correlation coefficient ≈ 0.88.
- Process Temperature was considered redundant.

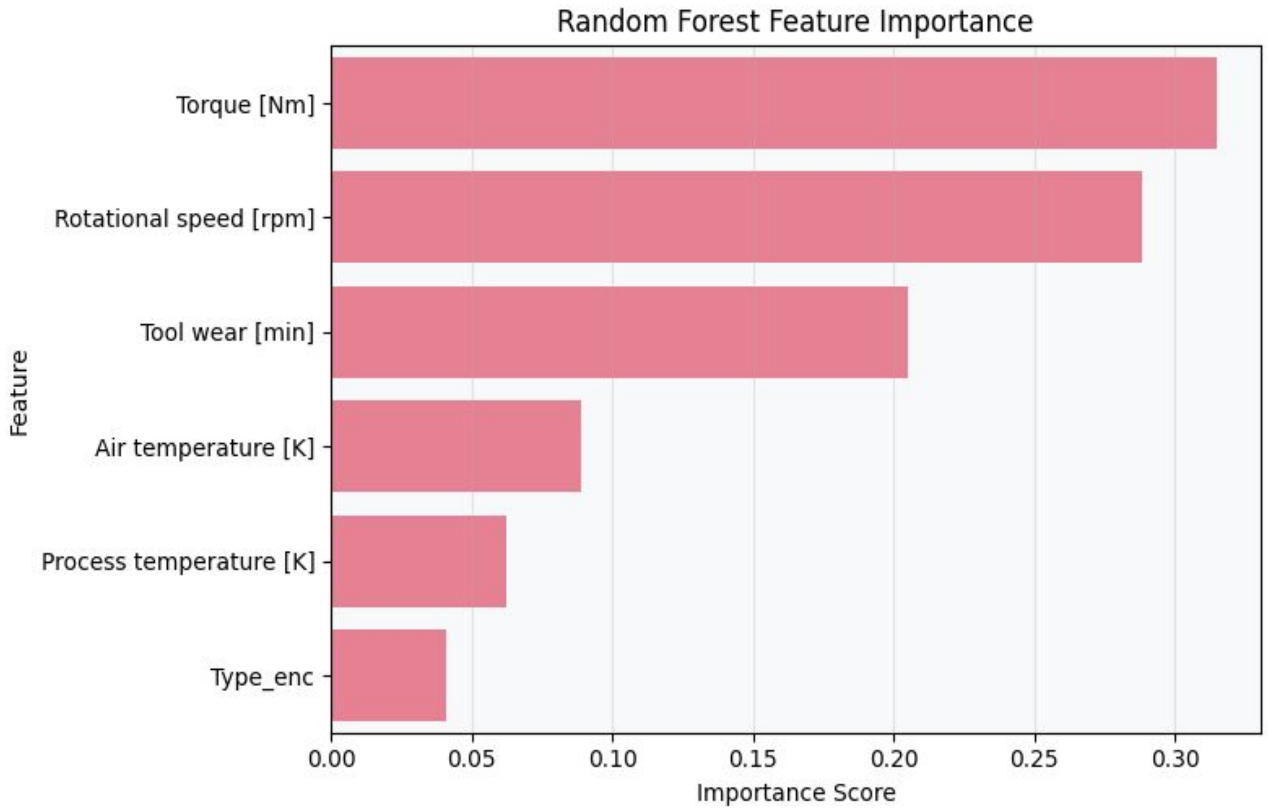


Feature Engineering Strategy

Random Forest Feature Importance

Observation

- Torque had the highest importance.
- Rotational Speed and Tool Wear were highly informative.
- Type had the lowest importance.
- Type was removed

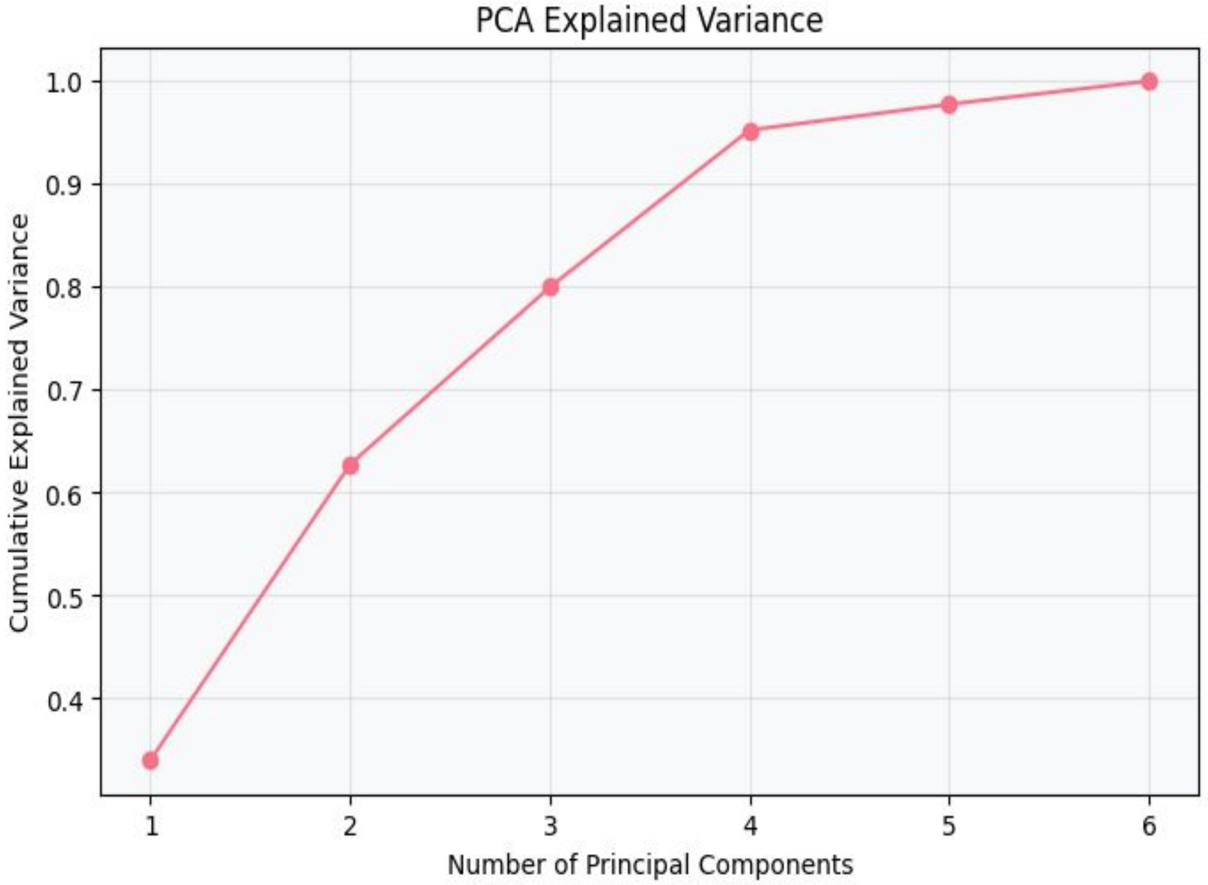


Feature Engineering Strategy

PCA Analysis

Observation

- PCA confirmed that most information is contained in fewer dimensions.
- Feature reduction was investigated and validated



4 Principal Components ≈95% explained variance

6 Principal Components ≈ 100% explained variance

Feature Engineering Strategy

Final Selected Features

Removed

- Type
- Process Temperature

Reason

- Low importance
- High correlation
- PCA verification

Stage	Features
Original	6
Reduced	4

Feature Engineering Evaluation

Original Features Only

```
original_feature_cols = [
    'Type_enc',
    'Air temperature [K]',
    'Process temperature [K]',
    'Rotational speed [rpm]',
    'Torque [Nm]',
    'Tool wear [min]'
]
```

Important Features

```
reduced_feature_cols = [
    'Air temperature [K]',
    'Rotational speed [rpm]',
    'Torque [Nm]',
    'Tool wear [min]'
]
```

Preprocessing Pipeline

1. Encoding
Converting 'Type' to numeric (L:1, M:2, H:0).

2. Balancing
Applying SMOTE to generate synthetic failure samples.

3. Scaling
Standardizing all features to Mean 0, Std 1.

Handling Class Imbalance

Synthetic Minority Over-sampling Technique (SMOTE):

Without balancing, models tend to predict "No Failure" for everything, yielding high accuracy but 0% recall for failures.

- **Before:** 9,661 (Non-Fail) vs 339 (Fail)
- **After SMOTE:** 9,661 (Non-Fail) vs 9,661 (Fail)
- **Result:** Balanced dataset with 19,322 total samples.

Standardization (Scaling)

Sensor values have vastly different ranges (e.g., Speed ~1500 vs Torque ~40). We use **StandardScaler**:

$z = (x - u) / s$

This is crucial for distance-based models (KNN, SVM) and gradient-descent based optimization.

Verification: After scaling, the training set mean is ~0.0000 for all features, ensuring numerical stability.

Data Splitting Strategy

Split applied to the resampled (SMOTE) balanced data:

Train (70%)

13,532 samples used for model learning.

Val (15%)

2,891 samples for hyperparameter tuning.

Test (15%)

2,899 samples for final evaluation.

Machine Learning Candidates

Seven different algorithms were evaluated to find the optimal PdM solution:

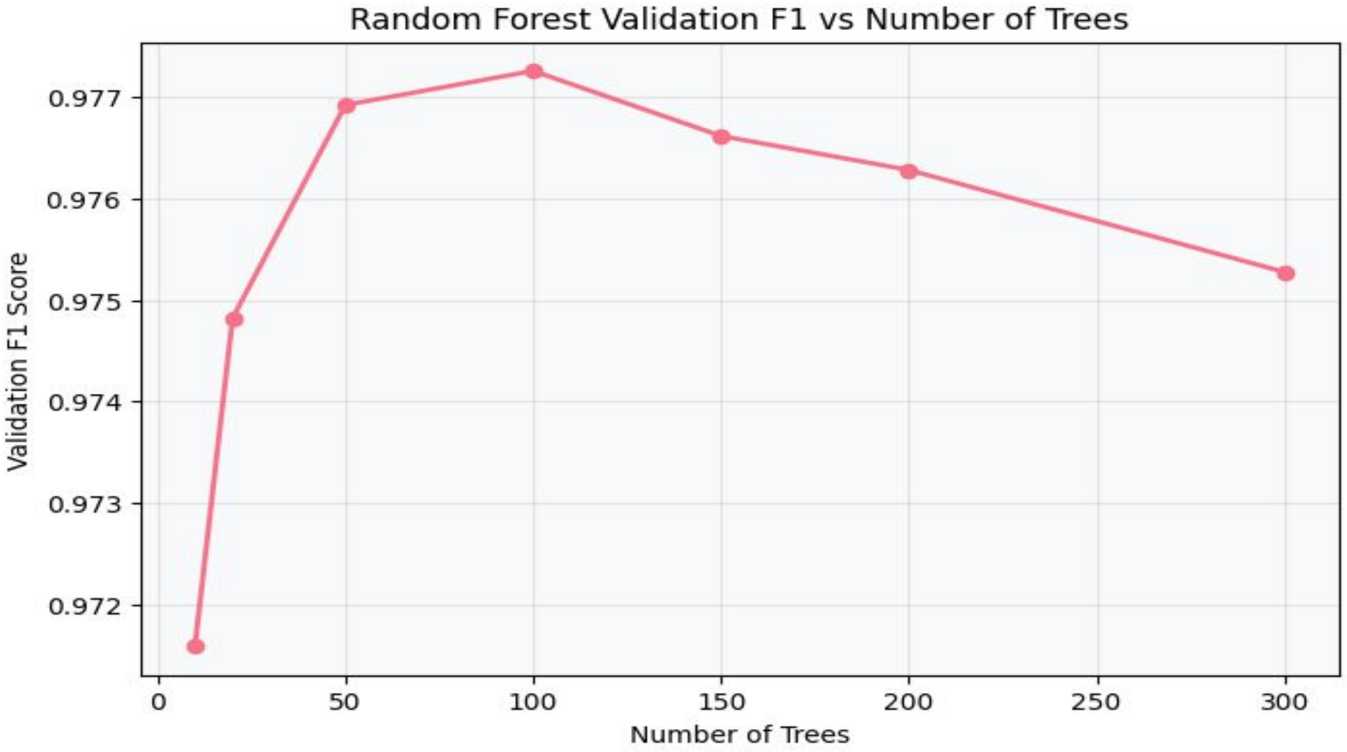
- **Linear:** Logistic Regression, SVM (RBF Kernel).
- **Trees:** Decision Tree, Random Forest.
- **Ensemble:** Gradient Boosting, AdaBoost.
- **Instance-Based:** K-Nearest Neighbors (KNN).

Section 4: Results & Business Impact

RANDOM FOREST ANALYSIS

Technologie Campus Cham

Random Forest Training:



Random Forest Test Results (useen Data):

Random Forest Results

	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.977171	0.977257	0.997648	0.974819	0.975179	0.997699
1	Reduced Features	0.967485	0.967896	0.993481	0.969645	0.970169	0.995972

LOGISTIC REGRESSION ANALYSIS

Logistic Regression Training:

Logistic Regression Test Results (useen Data):

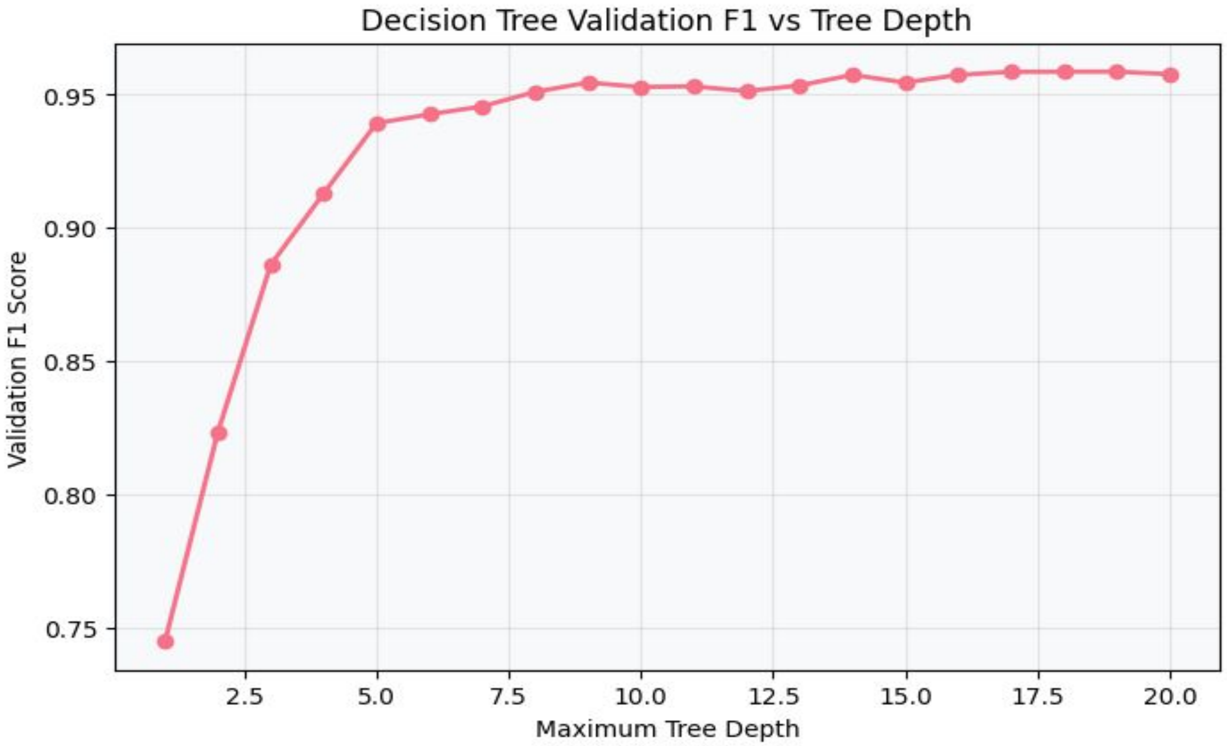
Logistic Regression Results

	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.828087	0.828562	0.913918	0.835805	0.834607	0.915351
1	Reduced Features	0.812522	0.811281	0.897780	0.813039	0.812327	0.897673

DECISION TREE ANALYSIS

Decision Tree

Training:



Decision Tree Test Results (useen Data):

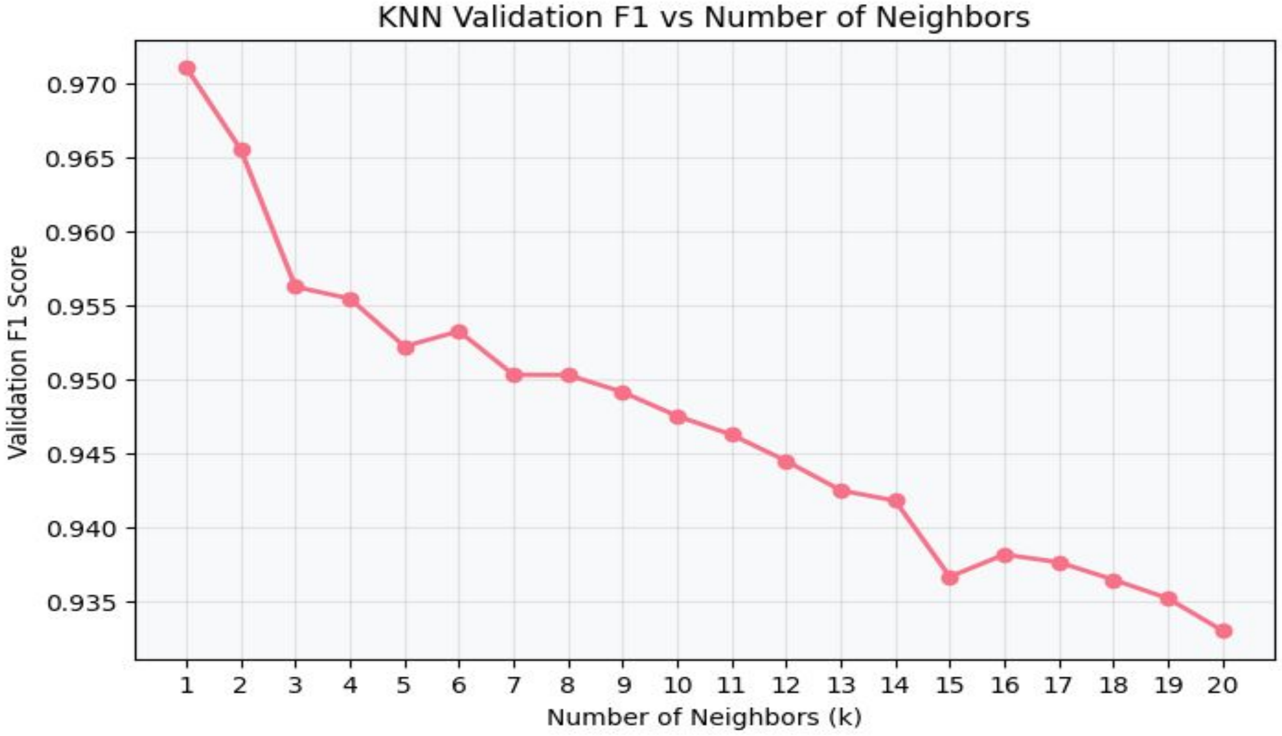
Decision Tree Results

	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.958492	0.958706	0.958490	0.962056	0.962329	0.962058
1	Reduced Features	0.952266	0.952675	0.952263	0.954467	0.954608	0.954468

K-NEAREST NEIGHBORS ANALYSIS

TECHNISCHE HOCHSCHULE DEGGENDORF
Technologie Campus Cham

KNN Training:



k = 1 gives the highest Validation F1.

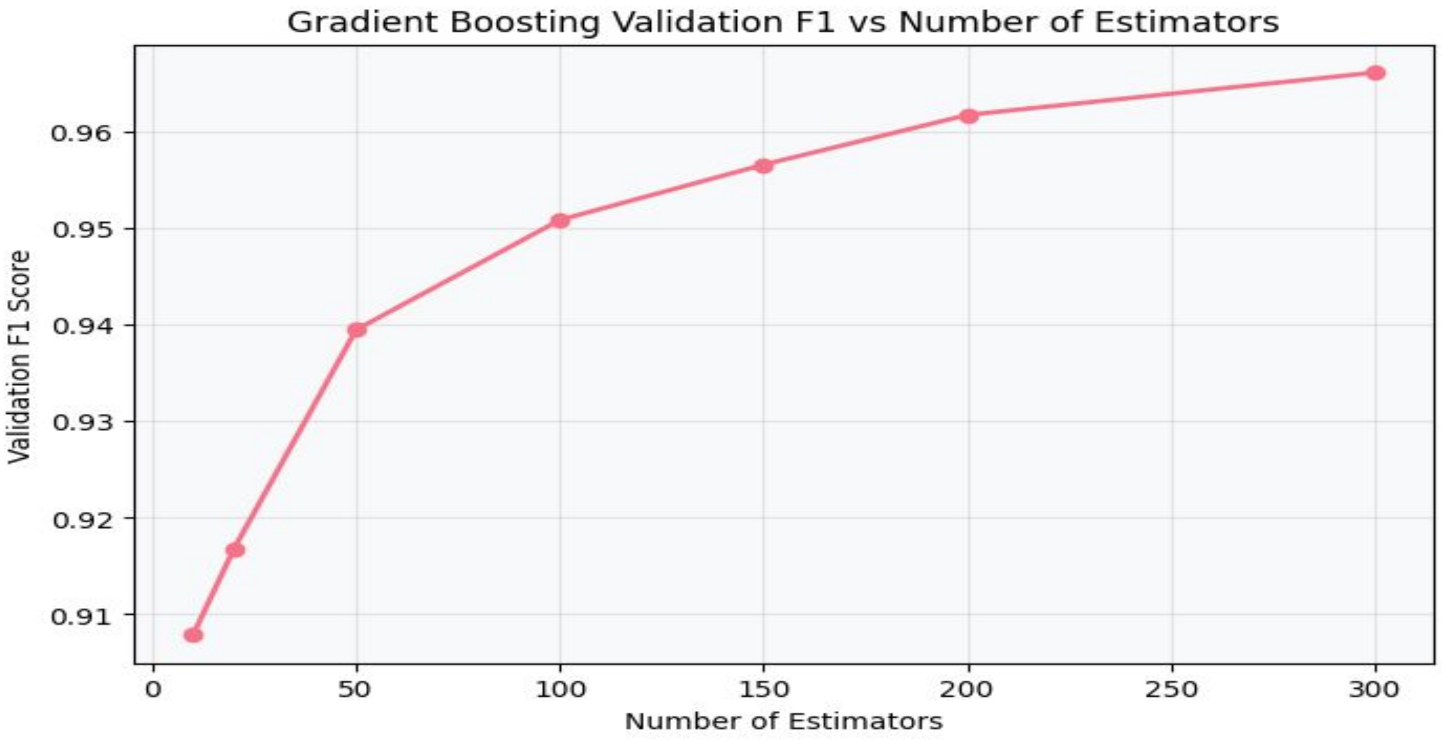
KNN Test Results (useen Data):

K-Nearest Neighbors Results

	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.950882	0.952221	0.982382	0.955157	0.956551	0.981935
1	Reduced Features	0.942926	0.944982	0.974853	0.941359	0.943409	0.977053

GRADIENT BOOSTING ANALYSIS

GRADIENT BOOSTING ANALYSIS Training:



GRADIENT BOOSTING ANALYSISTest Results (useen Data):

Gradient Boosting Results

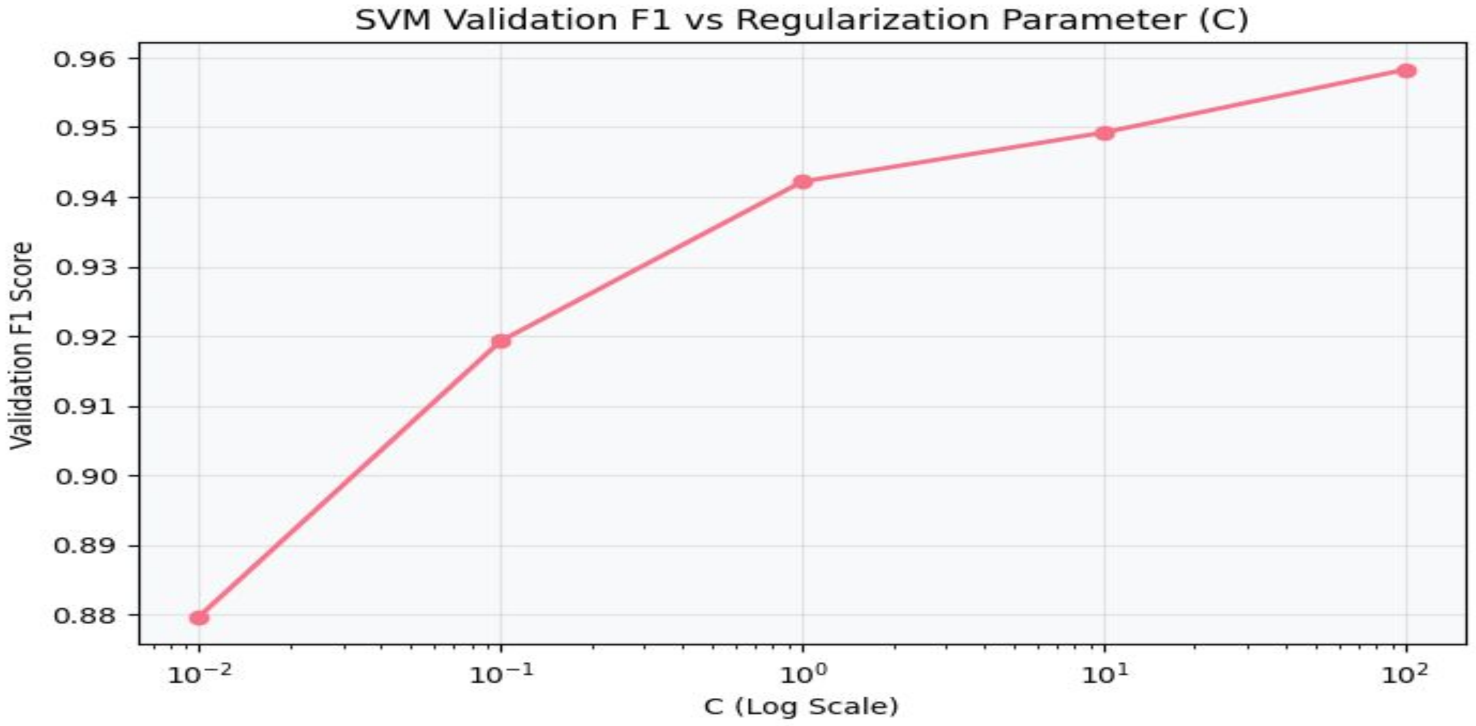
	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.950536	0.950876	0.988391	0.946188	0.947333	0.987038
1	Reduced Features	0.939813	0.940816	0.982396	0.940669	0.942127	0.981356

SUPPORT VECTOR MACHINE ANALYSIS

TECHNISCHE HOCHSCHULE DEGGENDORF
Technologie Campus Cham

SVM ANALYSIS

Training:



SVM Analysis Test Results (useen Data):

Support Vector Machine (RBF) Results

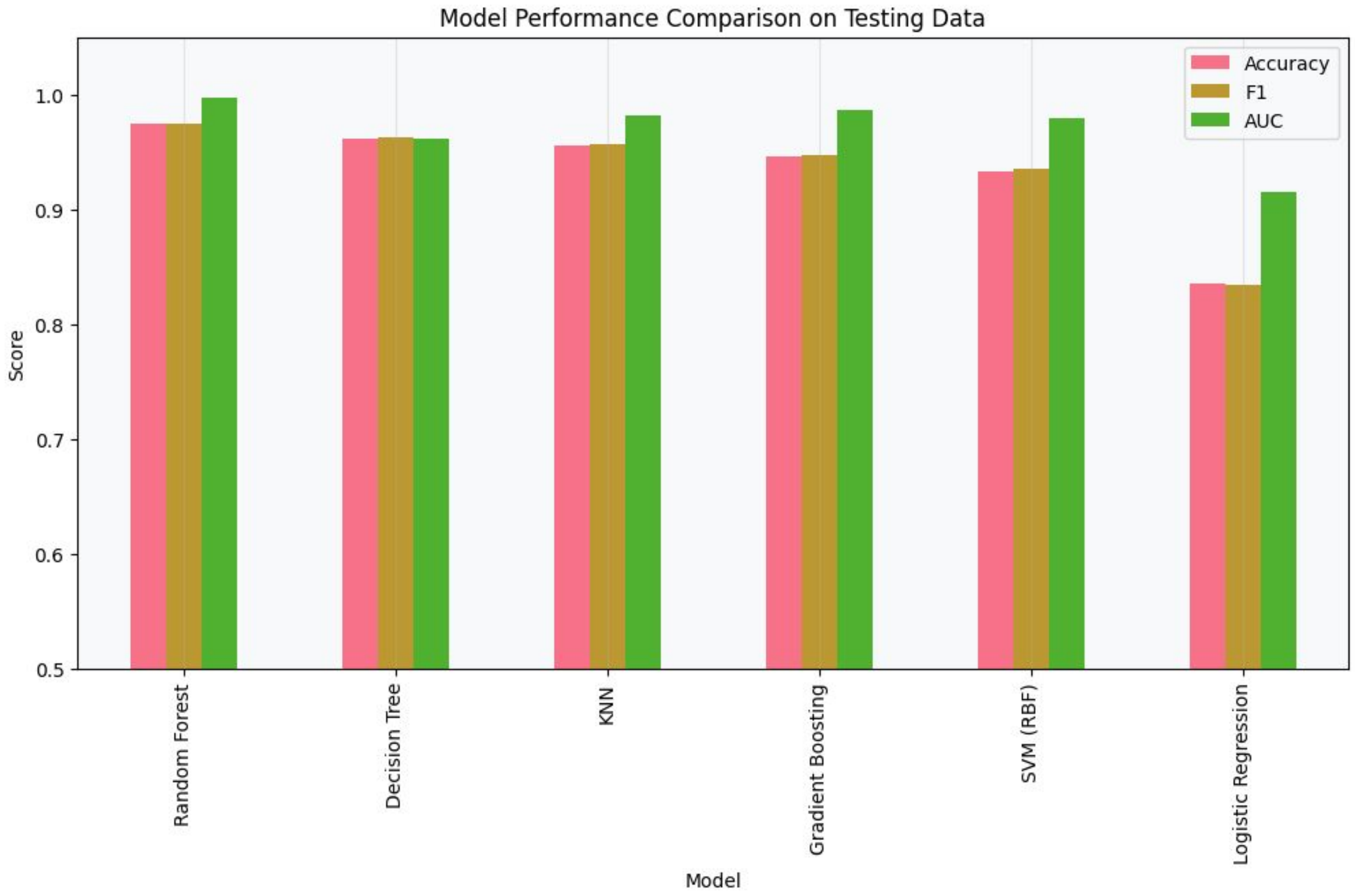
	Dataset	Validation Accuracy	Validation F1	Validation AUC	Test Accuracy	Test F1	Test AUC
0	Original Features	0.941197	0.942177	0.982734	0.933425	0.934995	0.978975
1	Reduced Features	0.922172	0.924674	0.971852	0.920662	0.923740	0.968953

All Models Comparison

Sorted by F1 Score (Highest to Lowest) on Test Data which is 15% of Dataset:

Model	Accuracy	Precision	Recall	F1	AUC
Random Forest	0.974819	0.961126	0.989648	0.975179	0.997699
Decision Tree	0.962056	0.955133	0.969634	0.962329	0.962058
KNN	0.955157	0.927414	0.987578	0.956551	0.981935
Gradient Boosting	0.946188	0.927297	0.968254	0.947333	0.987038
SVM (RBF)	0.933425	0.913158	0.957902	0.934995	0.978975
Logistic Regression	0.835805	0.840448	0.828847	0.834607	0.915351

All Models Comparison



Top Performer: Random Forest

The **Random Forest Classifier** achieved the best overall results across all metrics:

- **Accuracy:** 97.48%
- **Recall:** 98.96%
- **F1-Score:** 0.9752
- **AUC-ROC:** 0.9977

Why it won: By combining 100 decorrelated decision trees, it effectively captured the complex thermal-mechanical interactions while maintaining high robustness against noise.

Effect of Adding Features

Feature Engineering Evaluation

Original Features Only

```
original_feature_cols = [
    'Type_enc',
    'Air temperature [K]',
    'Process temperature [K]',
    'Rotational speed [rpm]',
    'Torque [Nm]',
    'Tool wear [min]'
]
```

Original + Engineered Features

```
#Prepare feature matrix and target
feature_cols = [
    'Type_enc',
    'Air temperature [K]',
    'Process temperature [K]',
    'Rotational speed [rpm]',
    'Torque [Nm]',
    'Tool wear [min]',
    'temp_diff',
    'power_W',
    'wear_rate',
    'torque_speed_ratio',
    'heat_stress'
]
```

Adding Feature Evaluation

The Random Forest model was trained twice to evaluate the impact of feature engineering. The engineered features were designed to capture machine stress, power consumption, and wear behavior.

=== RANDOM FOREST COMPARISON ===

	Without FE	With FE
Accuracy	0.9748	0.9796
Precision	0.9611	0.9734
Recall	0.9896	0.9862
F1-Score	0.9752	0.9798
AUC-ROC	0.9977	0.9982
MCC	0.9501	0.9594

Adding Feature Evaluation

Impact of Feature Engineering on Random Forest

Positive findings:

Accuracy improved by **0.49%**

Precision improved by **1.28%**

F1-score improved by **0.47%**

MCC improved by **0.98%**

AUC-ROC improved by **0.05%**

Feature engineering provided a measurable improvement in Random Forest performance. The increase in Precision, F1-Score, MCC, and AUC-ROC indicates that the engineered features contributed additional information related to machine operating conditions and failure behavior. **but it is small improvement and add complexity so it didn’t important to do**

Business & Operational Impact

Cost Savings

Predicting failures before they happen prevents expensive collateral damage to machine components.

Production Continuity

Reducing downtime by scheduling repairs during planned breaks rather than emergency stops.

Resource Efficiency

Tools are replaced only when the model predicts failure, not on fixed (conservative) intervals.

Any Questions ?

Thank you for your attention.